

ML Based Integrated Solution for Smart Pepper Farming in Sri Lanka

Presented by 25-26J-479





Problem Statement and Stakeholders

Today's Reality

- \$ Unfair pricing:** Prices change unpredictably, farmers lose income
- ? Poor quality grading:** Only 1L cup density used and ignores mold, adulteration
- ! Late disease detection:** New farmers lack knowledge, damage is done before action
- ↘ Unknown yields:** Guesswork planning leads to wasted resources

The Impact

Farmers lose income even with good harvests. Exporters reject batches. Ceylon pepper's global reputation weakens.



Who Needs This?

Primary: Small-scale pepper farmers in Sri Lanka

Secondary: Exporters, buyers, Agricultural officers and regulators

Research Gap

Yield Prediction

- Limited research integrating plant image analysis, environmental sensor data and machine learning models into a deployable decision-support system.
- Lack explain ability making them difficult for farmers to interpret.

Disease Detection

- Traditional systems only detect the disease, not how much the leaf is infected.
- Many systems also fail in real field conditions due to lighting changes and overlapping leaves.

Quality Grading

- Inaccurate Grading
- No Integrated System
- Lost Opportunities

Price Prediction

- ML models act as “black boxes”, reducing trust.
- Blockchain adoption is limited in developing economies.
- Forecasts often lack actionable insights for farmers.



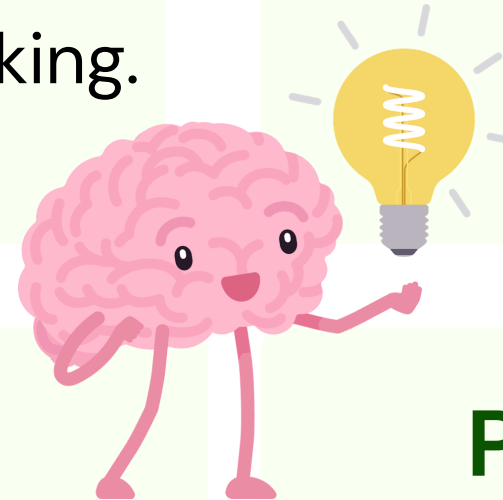
Research Objectives

Yield Prediction

To develop an intelligent crop yield prediction system for pepper cultivation using machine learning, image analysis and environmental data to support data-driven agricultural decision making.

Disease Detection

Develop an AI-driven system combining image processing and CNN deep learning that classifies black pepper leaf diseases



Quality Grading

To develop a pepper quality grading system using bulk density measurement, image analysis, and certification verification to accurately evaluate post-harvest pepper quality and support better decision-making for farmers and exporters.

Price Prediction & Blockchain

- Use ML model with multiple data sources to improve prediction accuracy.
- Combine market prices with blockchain in one system.
- Give simple “sell” or “wait” recommendations to farmers.

Solution Overview

Our solution combines yield prediction, price forecasting, quality grading, and disease detection into one platform to support pepper farmers and exporters.



Yield Prediction

ML-powered forecasting helps farmers plan resources, labor, and income accurately



Price Forecasting + Blockchain

ML predicts market prices; blockchain ensures transparent



Quality Grading

Image-based grading detects mold, adulteration, size variation, not just density

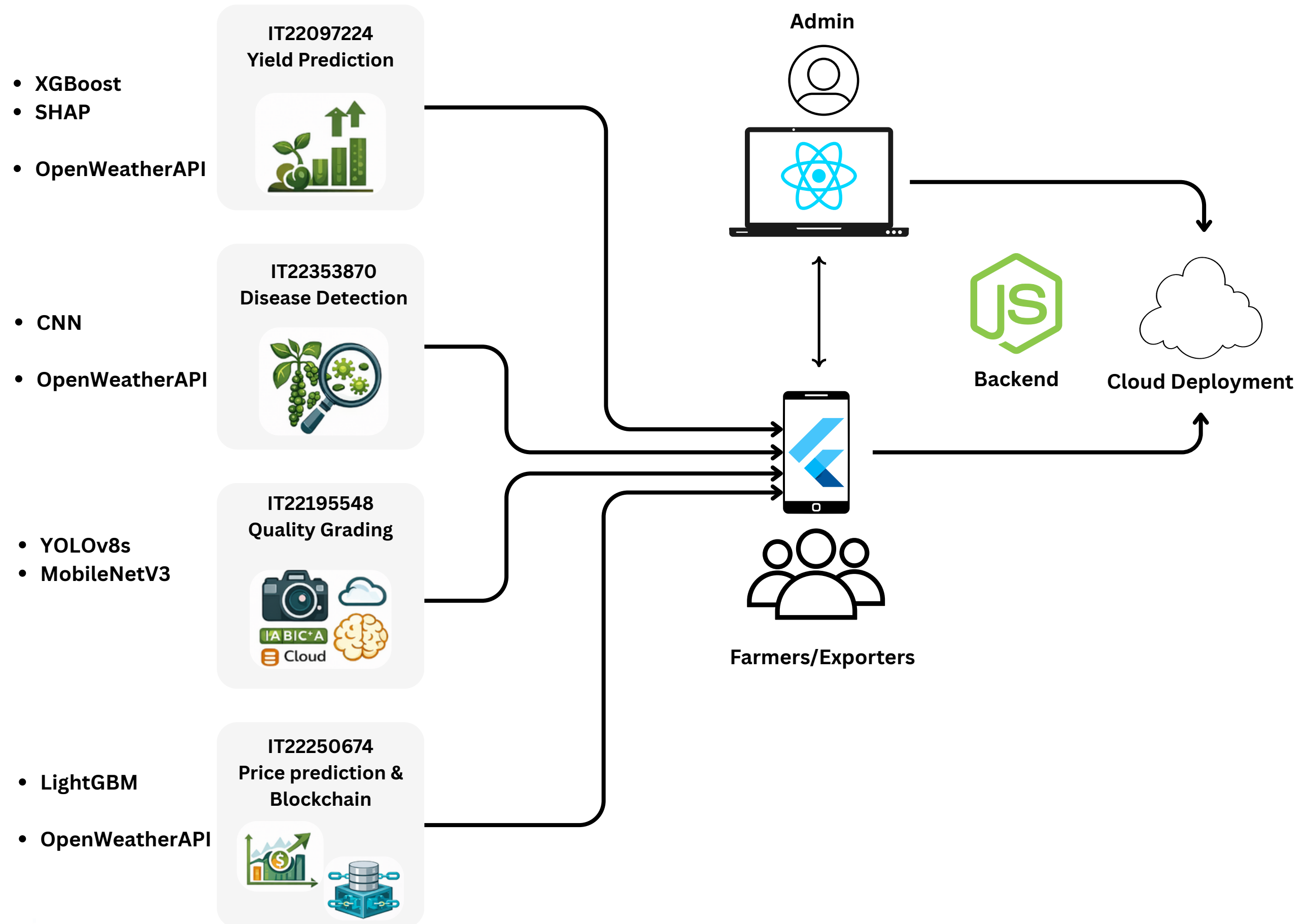


Disease Detection

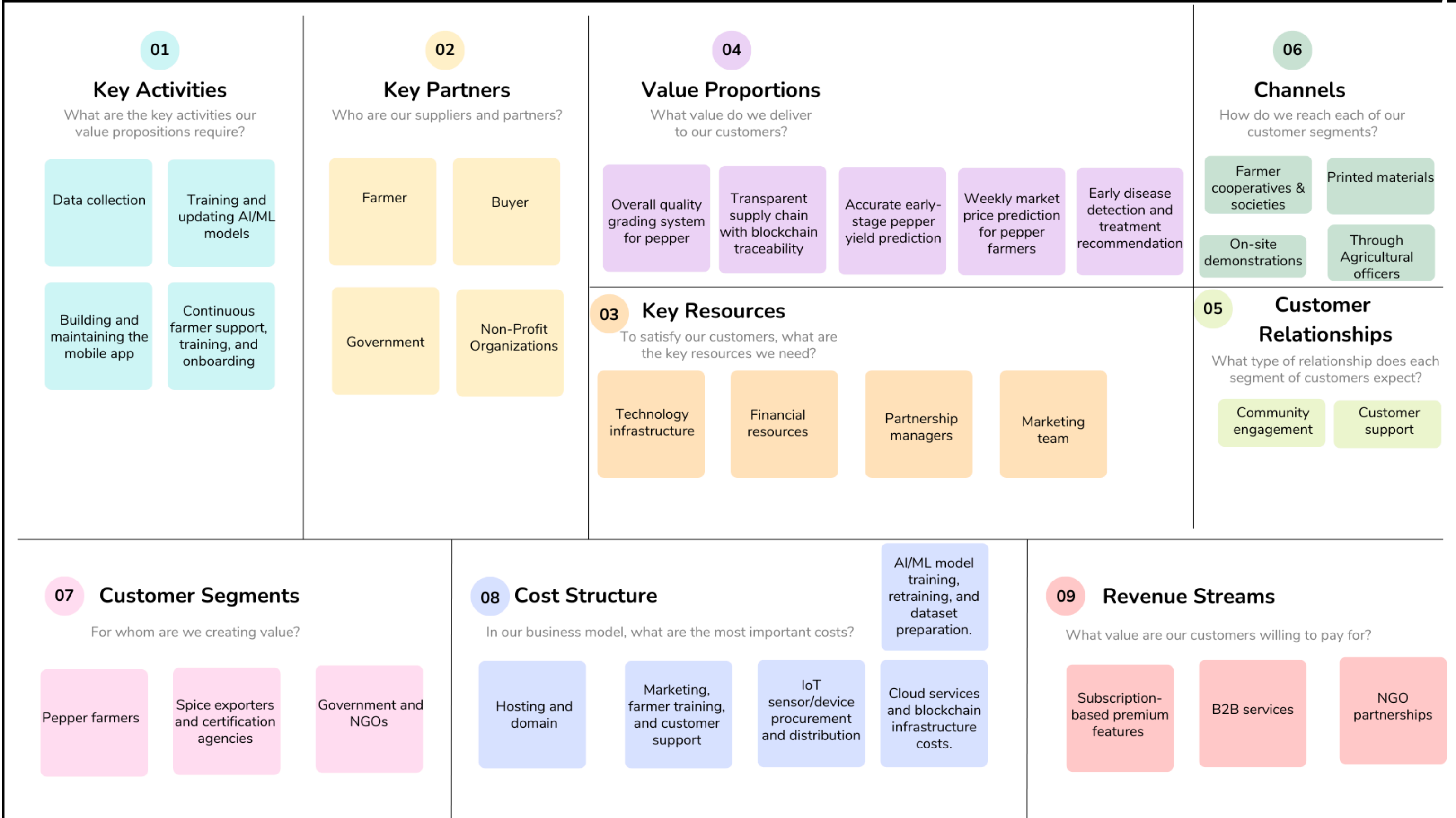
Early detection with infection spread percentage for timely intervention



System Architecture and Integration Diagram



Commercialization



Letter of Acceptance



February 27, 2026

TO WHOM IT MAY CONCERN

This is to confirm that a four-member final-year undergraduate research team from the Sri Lanka Institute of Information Technology (SLIIT), comprising **Ms. Madara Meegama, Ms. Janudi Adhikari, Ms. Ashika Ridimahaliyadda, and Mr. Induwara Ranasinghe**, presented to us the details of their proposed research project on the Sri Lankan pepper industry.

During the presentation, the team explained the objectives, methodology, and expected outcomes of their research, which focuses on developing a machine learning-based integrated solution for the Sri Lankan pepper industry. The students identified critical challenges within the sector, including unpredictable yields, inconsistent quality grading, lack of transparency in the supply chain, and limited access to advanced farming technologies for smallholder farmers. We reviewed their proposed approach and provided professional insights to enhance the quality, practicality, and applicability of their research.

The proposed solution integrates multiple components, including yield prediction supported by Explainable Artificial Intelligence (XAI), Internet of Things (IoT)-based post-harvest pepper quality grading, market price prediction supported by blockchain-based data integrity, and pepper leaf disease detection. We further supported the research team by facilitating interactions with farmers and industry stakeholders in the Kegalle District, enabling them to gather accurate field-level data and gain a better understanding of practical challenges. During our review, we identified several minor issues within certain components of their workflow and provided guidance and recommendations to refine and improve these areas.

Following the team's revisions, we reviewed the updated methodology and confirmed that the overall research process was technically sound, feasible, and suitable for practical application. In particular, we carefully evaluated the workflows proposed for the system's four main components: Yield Prediction, Pepper Quality Grading, Market Price Prediction, and Disease Detection.

We are satisfied with the revised methodologies and hereby endorse the overall research process and system design proposed by the team. We believe that this research has strong potential to deliver meaningful value to farmers, exporters, buyers, and the Sri Lankan agricultural sector as a whole.

We are pleased to **confirm that we supported this research initiative** and wish the team every success in their research endeavour.

A handwritten signature in blue ink, appearing to read "Tilak Karunaratne".

Tilak Karunaratne
Country Programme Manager
Oxfam Australia

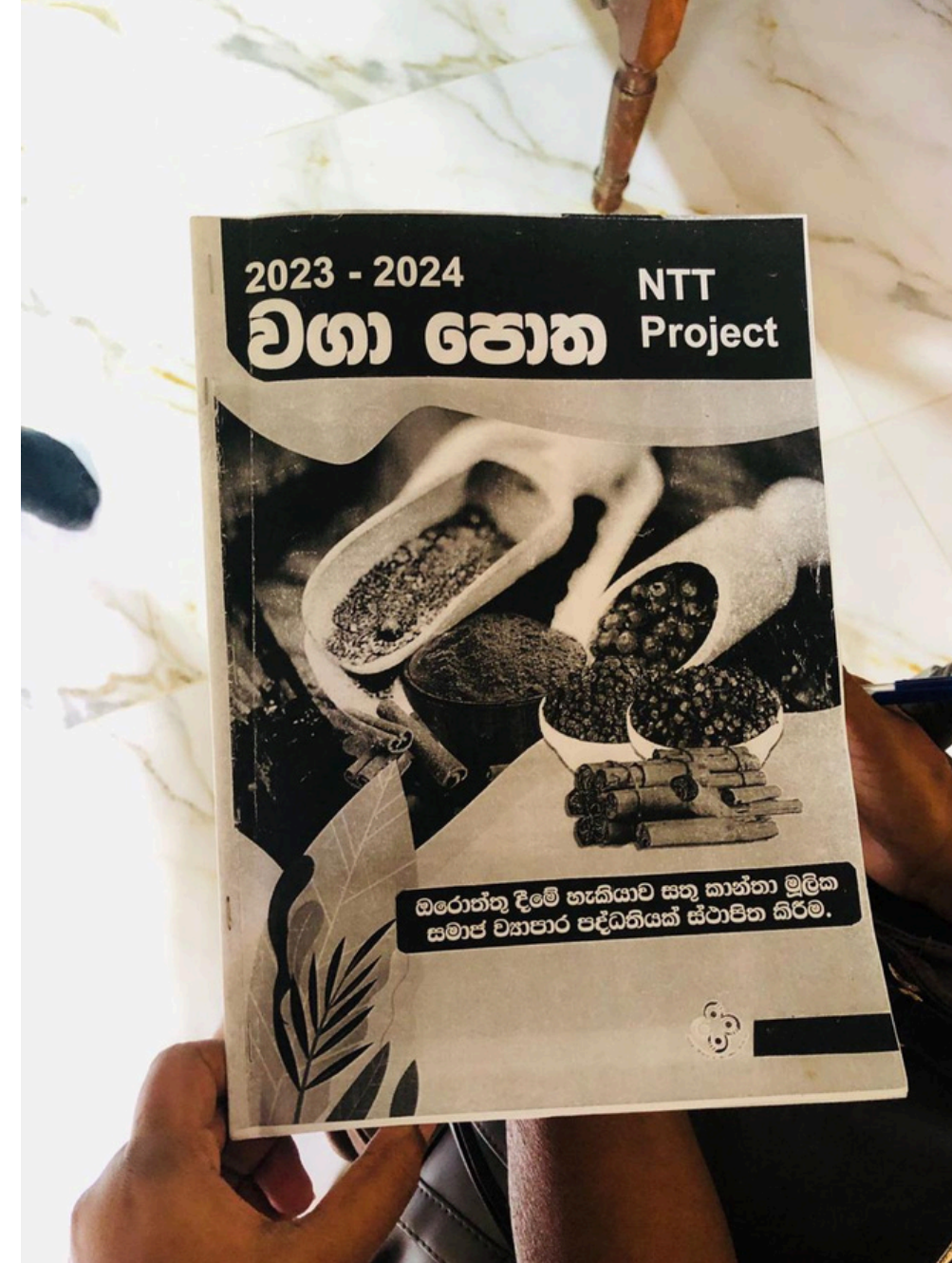
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Evidences



Evidences



Evidences

06. අස්වැන්න පෙළුම්

දිනය	ප්‍රථම වැන්න	ද්විතීයික වැන්න	තූල්‍යතාව (kg)	මිල	ආදායම්	වැය	ආදායම
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04. වගා බිමේ දෛනික සටහන්

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04/04/24	"	06:30 - 07:00	400	15000/-	25000/-	25000/-	25000/-

Evidences



Evidences



Demonstration



Thank You

